Amel Derras-Chouk

aderras.github.io
ad4429@columbia.edu

Education

PhD in Physics at CUNY Graduate Center, New York, NY — 2016–2022

Dissertation: *Collapse of Two-Dimensional Magnetic Skyrmions*

B.S. in Physics and Mechanical Engineering at Tufts University, Medford, MA — 2011–2015

Manuscripts in Prep

Derras-Chouk, Amel, Gregory Elsaesser, Zhengzhao Johnny Luo, Toshi Matsui, Andreas F. Prein, and Jingbo Wu (Accepted pending revision). *Estimating Vertical Velocity from Temperature, Pressure, and Latent Heating.*

Derras-Chouk, Amel, Zhengzhao Johnny Luo, and Gregory Elsaesser (In prep). *A Satellite-based Estimate of Deep Convective Precipitation Efficiency in the Tropics.*

Publications

Braneon, Christian, Luis Ortiz, Daniel Bader, Naresh Devineni, Philip Orton, Bernice Rosenzweig, Timon McPhearson, Lauren Smalls-Mantey, Vivien Gornitz, Talea Mayo, Sanketa Kadam, Hadia Sheerazi, Equisha Glenn, Liv Yoon, Amel Derras-Chouk, Joel Towers, Robin Leichenko, Deborah Balk, Peter Marcotullio, and Radley Horton (2024). “NPCC4: New York City climate risk information 2022—observations and projections”. In: *Annals of the New York Academy of Sciences* 1539.1, pp. 13–48.

Derras-Chouk, Amel and Zhengzhao Johnny Luo (2024). “A geostationary satellite-based approach to estimate convective mass flux and revisit the hot tower hypothesis”. In: *Surveys in Geophysics* 45.6, pp. 1959–1977.

Derras-Chouk, Amel, Eugene M Chudnovsky, and Dmitry A Garanin (2022). “Dynamics of the collapse of a ferromagnetic skyrmion in a centrosymmetric lattice”. In: *Physical Review B* 105.13, p. 134432.

Derras-Chouk, Amel and Eugene M Chudnovsky (2021). “Skyrmions near defects”. In: *Journal of Physics: Condensed Matter* 33.19, p. 195802.

Derras-Chouk, Amel, Eugene M Chudnovsky, and Dmitry A Garanin (2021). “Quantum states of a skyrmion in a two-dimensional antiferromagnet”. In: *Physical Review B* 103.22, p. 224423.

– (2019). “Thermal collapse of a skyrmion”. In: *Journal of Applied Physics* 126.8.

– (2018). “Quantum collapse of a magnetic skyrmion”. In: *Physical Review B* 98.2, p. 024423.

Derras-Chouk, Amel, Eugene M Chudnovsky, Dmitry A Garanin, and Reem Jaafar (2018). “Graphene cantilever under Casimir force”. In: *Journal of Physics D: Applied Physics* 51.19, p. 195301.

Research Experience

Postdoctoral Researcher, Columbia Climate School — *Sept. 2024–present*

Postdoctoral Researcher, The City College of New York, CUNY — *Sept. 2022–Aug. 2024*

Graduate Student Intern, NASA Goddard Institute for Space Studies — *Oct. 2021–Aug. 2022*

Graduate Student Researcher, CUNY Graduate Center — *July 2017–Sept. 2022*

Teaching Experience

Ajunct Lecturer Lehman College — *Aug. 2017–May 2021*

– Lab Section of General Physics 1 (PHY 166/168)

– Lab Section of General Physics 2 (PHY 167/169)

Selected Presentations

Jan. 2026 American Meteorological Society Annual Meeting 2026, Houston, TX, USA. *Estimating vertical velocity from temperature, pressure, and latent heating*

Dec. 2025 American Geophysical Union Annual Meeting, New Orleans, LA, USA. *The diurnal cycle of deep convective precipitation efficiency in the tropics*

Nov. 2025 Invited talk at INCUS Science Team Meeting, Fort Collins, CO, USA. *Towards Developing a Historical Record of Updraft Vertical Velocity*

Oct. 2025 Columbia Climate School Research Symposium, New York, NY, USA. *How fast are thunderstorm updrafts?*

Sept. 2025 Invited seminar at NOAA Geophysical Fluid Dynamics Laboratory, Princeton, NJ, USA. *Quantifying Vertical Motion Using Satellite Data.*

July 2025 Gordon Research Conference on Atmospheric Radiation (Poster presentation), Lewiston, ME, USA. *Estimating vertical velocity from temperature, pressure, and latent heating.*

Apr. 2025 NASA Goddard Institute for Space Science, New York, NY, USA. *Quantifying Convective Vertical Velocity and Precipitation Efficiency Using Satellite Data.*

Feb. 2025 CloudSat Calipso Science Team Meeting, Fort Collins, CO, USA. *Estimating convective mass flux, detrainment, and precipitation efficiency using CloudSat, Geostationary, and IMERG data.*

Dec. 2023 AGU Annual meeting (Poster presentation), San Francisco, CA. *A Satellite-Based Approach to Estimating Convective Mass Flux and Revisiting the Hot Tower Hypothesis*

Community Service

Volunteer Word Up Community Bookshop/Libreria Communitaria *Sept. 2017–Dec. 2023*

Honors and Awards

CUNY Graduate Center Dissertation Fellowship 2020–2021

Physics Tithe Fellowship 2018–2022

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